

Cloud Adoption & Migration : Experiment - 7

Aim:

To analyze and compare the environmental impact of carbon-neutral cloud services with traditional (non-carbon-optimized) cloud infrastructure, using official carbon footprint tools and sustainability reports provided by cloud service providers.

Theory:

Traditional data centers and early cloud infrastructures depend largely on grid electricity, which often includes fossil fuel-based energy sources, leading to increased carbon emissions and environmental impact. With the growing need for sustainable computing, modern cloud providers have introduced carbon-neutral and carbon-aware services that utilize renewable energy and energy-efficient infrastructure. These advancements aim to reduce the overall carbon footprint of cloud operations. By leveraging sustainability tools and optimized deployment strategies, organizations can make environmentally responsible decisions. This experiment focuses on analyzing and comparing these approaches to highlight the benefits of carbon-neutral cloud computing.

Implementation Steps:

1. Standard Workload:

Component	Specification
Compute	1 Virtual Machine (2 vCPU, 4 GB RAM)
Storage	100 GB object storage
Database	Small managed database
Network	Moderate monthly traffic
Usage	30 Days

2. Traditional Cloud Deployment Scenario:

- a. Region: Any default region (e.g., Asia Pacific / Mumbai / US East)
- b. No focus on carbon emissions
- c. Standard VM + storage services
- d. No renewable energy consideration

The workload is deployed in a standard cloud region without considering carbon efficiency or sustainability metrics.

3. Carbon-Neutral Cloud Scenario:

In this scenario, the given workload is deployed in a sustainable and environmentally optimized cloud environment. The focus is on reducing carbon emissions by selecting regions and services that prioritize renewable energy usage. The deployment is carried out in a low-carbon intensity region, where cloud data centers are powered predominantly by renewable energy sources such as solar, wind, or hydroelectric power. These regions are specifically designed to minimize environmental impact and support carbon-neutral or carbon-free operations. Cloud providers offer dedicated sustainability tools and dashboards that help monitor and analyze carbon emissions. These tools provide insights into energy consumption, carbon footprint, and

region-wise carbon intensity, enabling better decision-making for environmentally responsible deployments.

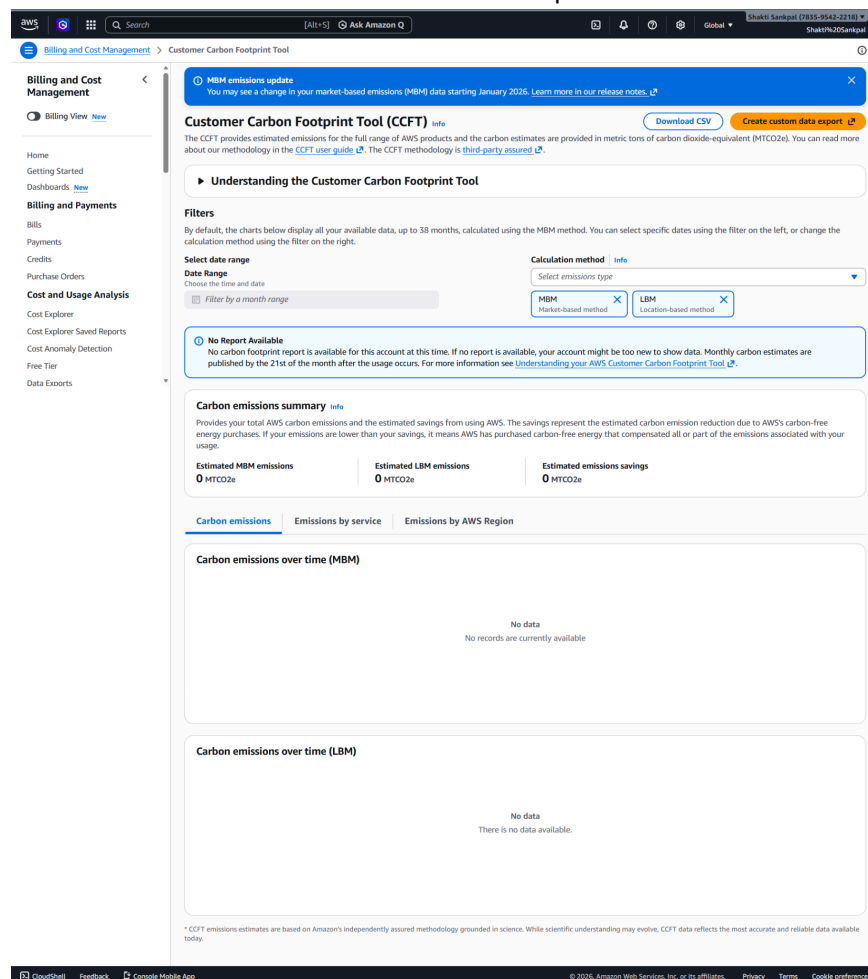
Examples of Carbon-Neutral Cloud Implementations:

- Google Cloud: Provides regions with high Carbon-Free Energy (CFE) percentage, ensuring most operations are powered by renewable energy.
- Amazon Web Services: Offers regions supported by renewable energy projects and aims for 100% renewable energy usage.
- Microsoft Azure: Provides sustainability reporting tools and operates energy-efficient, low-carbon data centers.

4. Analysis Using AWS Customer Carbon Footprint Tool

In this step, the Amazon Web Services Customer Carbon Footprint Tool is used to estimate and analyze the carbon emissions generated by the defined workload.

- Logged into the AWS Management Console.
- Accessed the Customer Carbon Footprint Tool from the services menu.
- Observed carbon emission data including total CO₂ emissions and service-wise distribution.
- Analyzed emission trends and estimated environmental impact.
- Compared a traditional deployment region with a sustainable region backed by renewable energy.
- Recorded observations in tabular form and captured dashboard screenshots.



The dashboard displayed the message: “No carbon footprint report is available for this account at this time.” This indicates that the AWS account currently has no recorded resource usage, and therefore, no carbon emissions data is generated.

The AWS Customer Carbon Footprint Tool calculates emissions based on actual resource usage such as virtual machines, storage, and network activity. Since no services are actively used in this account, the tool does not generate emission data.

Assumed Analysis:

Scenerio	Carbon emissions	Energy Sources
Traditional Cloud	Higher	Grid Based Energy
Carbon Neutral Cloud	Lower	Renewable Energy

5. Comparison Between Traditional Cloud and Carbon-Neutral Cloud

In this step, a comparative analysis is performed between a traditional cloud deployment and a carbon-neutral cloud deployment based on various sustainability parameters. This comparison highlights the environmental impact and efficiency of both approaches.

Parameter	Traditional Cloud	Carbon-Neutral Cloud
Energy Source	Grid-based energy (fossil fuels)	Renewable energy (solar, wind, hydro)
Carbon Emissions	Higher	Lower
Region Selection	No sustainability consideration	Low-carbon / optimized regions
Transparency	Limited visibility of emissions	High visibility using dashboards
Sustainability Tools	Not actively used	Actively used (carbon footprint tools)
Environmental Impact	Higher carbon footprint	Reduced carbon footprint
Long-term Impact	Not sustainable	Supports sustainability goals

The comparison clearly shows that traditional cloud deployments rely heavily on conventional energy sources, resulting in higher carbon emissions. In contrast, carbon-neutral cloud deployments utilize renewable energy and optimized regions, significantly reducing environmental impact.

Cloud providers such as Amazon Web Services, Google Cloud, and Microsoft Azure offer tools and infrastructure that support sustainable computing practices.

6. Interpretation and Discussion of Results

In this step, the results obtained from the comparative analysis are interpreted to understand the impact of cloud deployment strategies on carbon emissions and sustainability.

a. Reason for Difference in Emissions:

The difference in carbon emissions between traditional and carbon-neutral cloud environments is primarily due to the type of energy sources used by data centers. Traditional cloud regions rely on conventional grid electricity, which often includes fossil fuels such as coal and natural gas. This leads to higher carbon emissions. In contrast, carbon-neutral cloud regions are powered by renewable energy sources such as solar, wind, and hydroelectric power, which significantly reduce carbon emissions.

b. Role of Renewable Energy

Renewable energy plays a crucial role in minimizing the environmental impact of cloud computing. By using clean energy sources, cloud providers reduce greenhouse gas emissions and promote sustainable operations. Major providers like Amazon Web

Services, Google Cloud, and Microsoft Azure are investing heavily in renewable energy projects to power their data centers.

c. Importance of Region Selection

Region selection is a key factor in determining the carbon footprint of cloud workloads. Different regions have different energy mixes and carbon intensities. Deploying applications in regions with low carbon intensity can significantly reduce emissions without changing the workload itself. Therefore, choosing an appropriate region is an effective strategy for sustainable cloud computing.

d. Impact on Organizational Sustainability Goals

Using carbon-neutral cloud strategies helps organizations: Reduce their overall carbon footprint, Meet environmental, social, and governance (ESG) goals, Improve corporate social responsibility (CSR), Comply with environmental regulations
This also enhances the organization's reputation as an environmentally responsible entity.

Conclusion:

This experiment successfully analyzed and compared the environmental impact of traditional and carbon-neutral cloud deployments. It was observed that carbon-neutral cloud services significantly reduce emissions by utilizing renewable energy and optimized regions. The use of cloud sustainability tools enables better visibility and informed decision-making regarding carbon footprint. Therefore, adopting carbon-aware cloud strategies is essential for achieving sustainable and environmentally responsible computing.